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PIMPRI CHINCHWAD UNIVERSITY

Department of CSE - Artificial Intelligence & Data Science

Course Report

Mini Project Report

Subject: Data Science and Analytics

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Submitted By

Shravan Raut

Roll No: SOE23201030117

Department of AI & DS

Pimpri Chinchwad University

Guided By

Prof. Tushar R. Mahore

Assistant Professor, CSE - AI&DS

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Comprehensive Analysis of BMW Car Prices Using Exploratory Data Analysis and Linear Regression in R

1 Objective and Problem Statement

The objective of this project is to analyze a real-world BMW car dataset obtained from Kaggle, understand the statistical distribution of its features, and examine the impact of mileage on price. The project uses exploratory data analysis (EDA) and simple linear regression to model the relationship between mileage and price.

Problem Statement: Car buyers and sellers often rely on assumptions about depreciation due to mileage. This analysis aims to quantify how mileage affects BMW car prices and build a simple predictive model using R.

2 Dataset Description

2.1 Source

The dataset used, `bmw.csv`, was downloaded from Kaggle, a widely used platform for high-quality real-world datasets.

2.2 Dataset Size

- Number of rows: varies depending on final cleaned dataset
- Number of columns: typically around 8–10 attributes

2.3 Important Features Used

For this project, two key features were analyzed:

- **Price** – Selling price of the BMW car (numeric)
- **Mileage** – Distance the car has been driven (numeric)

2.4 Data Quality

Missing values were detected using R's `is.na()` function and removed using `na.omit()`. The dataset is clean and suitable for analysis.

3 Methodology

3.1 Step 1: Data Import

The dataset was imported using:

```
bmw_data <- read.csv("bmw.csv", stringsAsFactors = FALSE)
```

3.2 Step 2: Data Exploration

Initial data understanding was performed using:

```
head(bmw_data)
str(bmw_data)
summary(bmw_data)
```

3.3 Step 3: Data Cleaning

Missing values were counted and removed:

```
colSums(is.na(bmw_data))
bmw_data <- na.omit(bmw_data)
```

3.4 Step 4: Exploratory Data Analysis

Two major visualizations were created:

Price Distribution

```
# Price distribution histogram  
hist(bmw_data$price, main="Distribution of BMW Car Prices",  
      xlab="Price", col="lightblue", breaks=20)
```

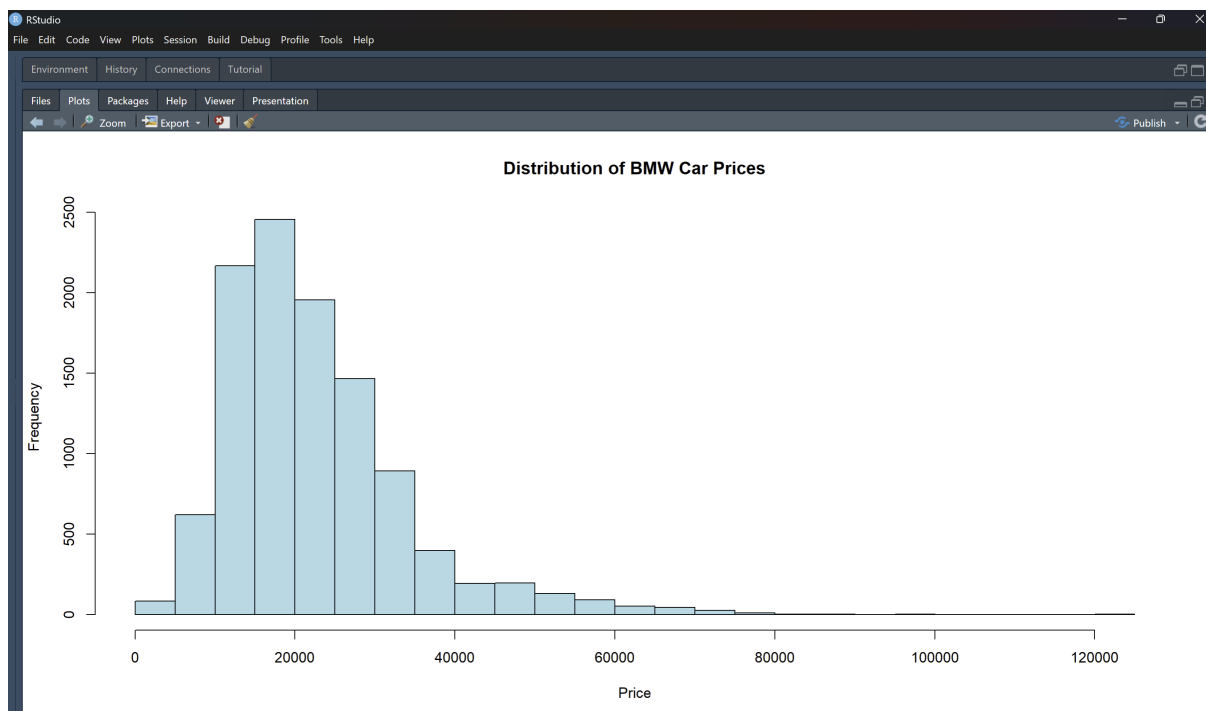


Figure 1: Histogram: Distribution of BMW Car Prices

Price vs Mileage

```
# Relationship between price and mileage scatter plot  
plot(bmw_data$mileage, bmw_data$price,  
      main="Price vs Mileage",  
      xlab="Mileage", ylab="Price",  
      pch=20, col="blue")
```



Figure 2: Scatter Plot: Price vs Mileage

3.5 Step 5: Linear Regression Modeling

A simple regression model was built:

```
lm_model <- lm(price ~ mileage, data = bmw_data)
# Plot regression line
plot(bmw_data$mileage, bmw_data$price,
     main="Price vs Mileage with Regression Line",
     xlab="Mileage", ylab="Price", pch=20, col="blue")
abline(lm_model, col="red", lwd=2)
```

A regression line was plotted:

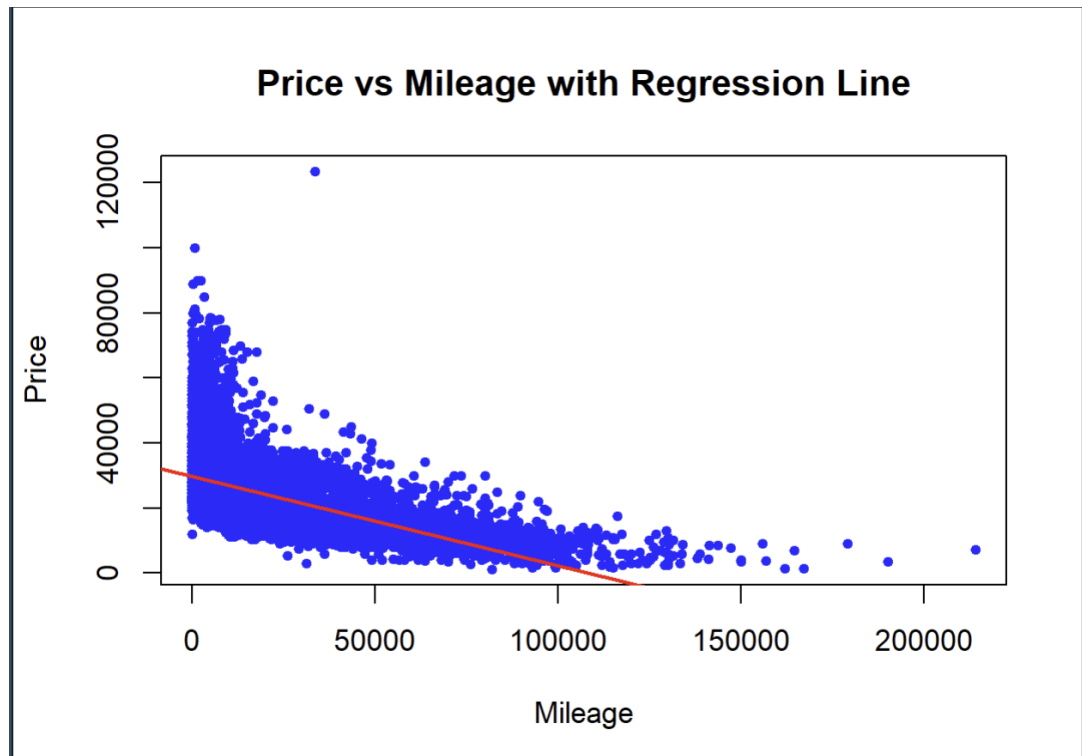


Figure 3: Price vs Mileage with Regression Line

3.6 Step 6: Model Evaluation

Predicted values and residuals were generated:

```
predicted <- lm_model$fitted.values
residuals <- lm_model$residuals
```

Actual vs Predicted Table

Index	Actual Price	Predicted Price
1	11200	11307
2	27000	25666
3	16000	16000
4	12750	12750

Table 1: Actual vs Predicted Price (First 10 Records)

4 Results

4.1 Statistical Findings

- Cars with higher mileage generally have a lower price.
- The histogram indicates that most BMW cars fall in the mid-price range.
- The scatter plot shows a negative correlation between mileage and price.

4.2 Model Insights

The linear regression model shows a clear downward trend:

$$\text{Price} = \beta_0 + \beta_1 \times \text{Mileage}$$

where:

- β_1 is expected to be negative
- Mileage is a significant predictor of car price

5 Conclusion

The analysis concludes that mileage has a significant impact on the price of a BMW car. As mileage increases, the price tends to decrease. This aligns with real-world expectations of vehicle depreciation.

Linear regression provided a simple yet effective model to quantify this relationship. The visualizations helped understand overall data distribution and trends.

6 Future Scope

Future work can include:

- Adding more features such as model year, engine size, and fuel type.
- Building multiple regression models for better prediction accuracy.
- Applying machine learning models such as Random Forest
- Creating dashboards using Shiny for interactive analysis.